

Raccolta Risultati dei Notebook

Indice

1	liverDataSetColab.ipynb	2
1.1	Tabelle	2
1.2	Grafici	2
2	teenDataset.ipynb	4
2.1	Tabelle	4
2.2	Grafici	5
3	AdultIncomeDataset.ipynb	7
3.1	Tabelle	7
3.2	Grafici	7

1 liverDataSetColab.ipynb

1.1 Tabelle

Tabella 1: Mutual information raggruppata – Liver

Variabile	Importanza
Sgpt	0.081705
Age	0.068543
DB	0.063133
TB	0.055864
ALB	0.044244
Alkphos	0.036145
Sgot	0.028294
A/G Ratio	0.021321
Gender	0.013342
TP	0.000000

Tabella 2: Riepilogo robustezza NN e SVM – Liver

Modello	Gruppo	Acc 0	Bal 0	Score 0	Acc 50	Bal 50	Score 50	Acc 100	Bal 100	Score 100
NN	Peggiori	0.7280	0.7364	0.7019	0.7326	0.7228	0.6977	0.7280	0.7052	0.6862
NN	Migliori	0.7280	0.7364	0.7019	0.6926	0.7164	0.6710	0.6903	0.7244	0.6725
SVM	Peggiori	0.6229	0.7060	0.6184	0.6457	0.7160	0.6383	0.6229	0.7000	0.6173
SVM	Migliori	0.6229	0.7060	0.6184	0.6457	0.7220	0.6395	0.6229	0.7180	0.6201

Nota: per il Decision Tree nel notebook sono presenti i grafici finali e parte dell'output numerico, ma non un dataframe serializzato pulito e coerente come per NN e SVM; per questo nel riepilogo tabellare qui sopra ho riportato solo i modelli con tabella numerica completa salvata nel notebook.

1.2 Grafici

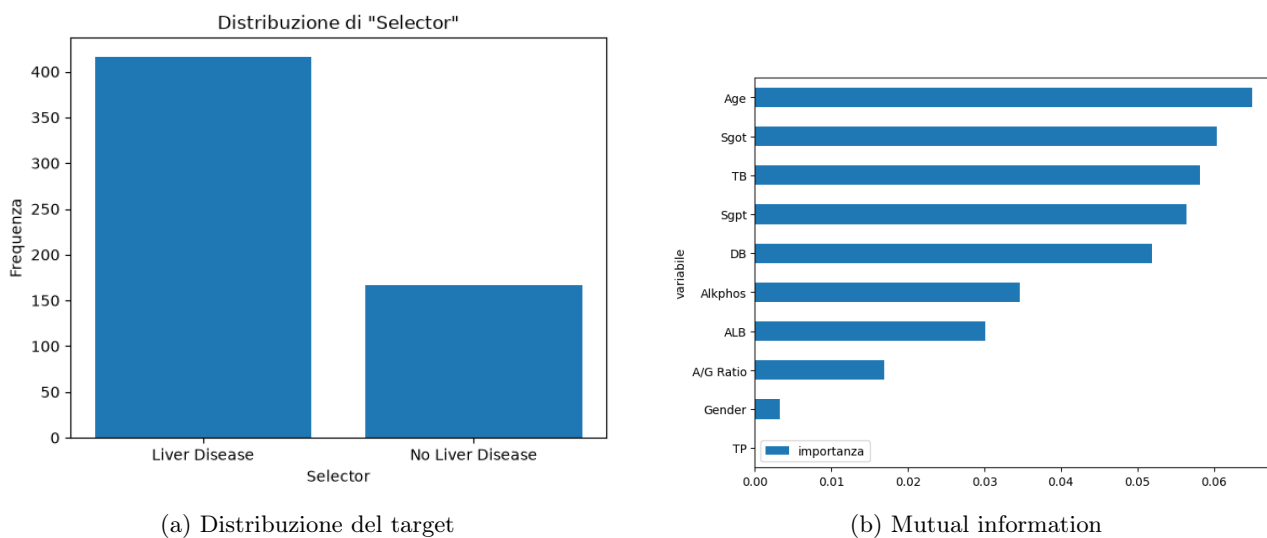
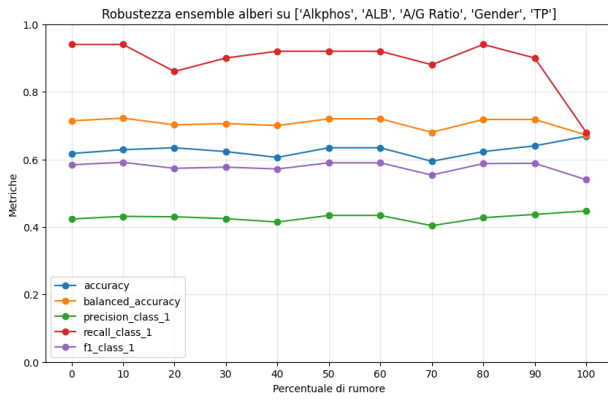
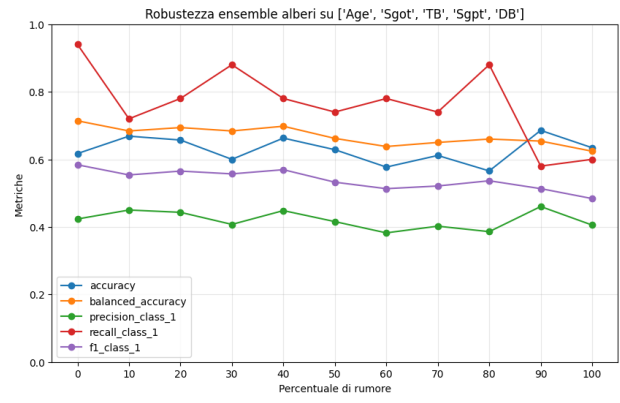


Figura 1: Panoramica dataset Liver.

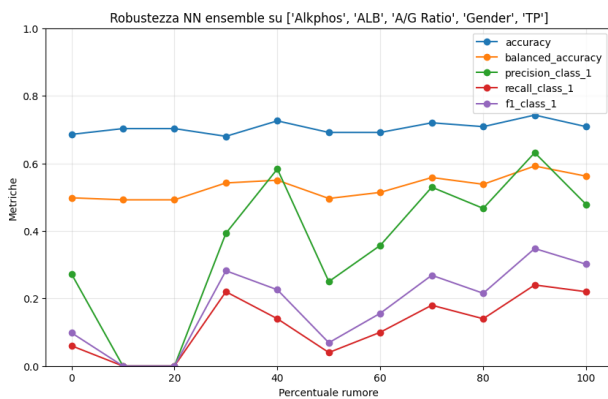


(a) Decision Tree, feature peggiori

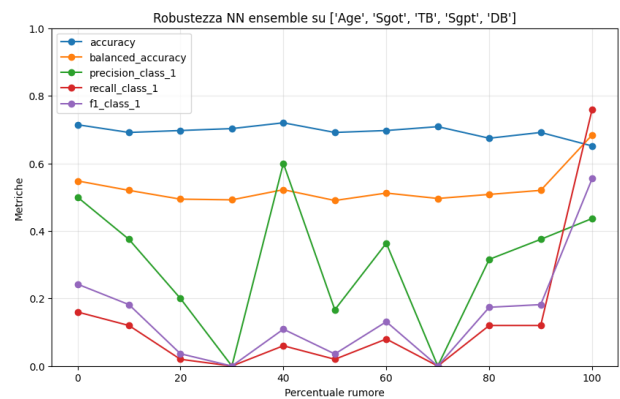


(b) Decision Tree, feature migliori

Figura 2: Robustezza Decision Tree – Liver.

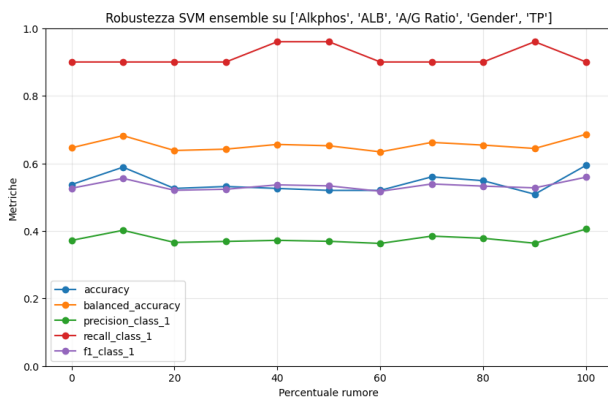


(a) Neural Network, feature peggiori

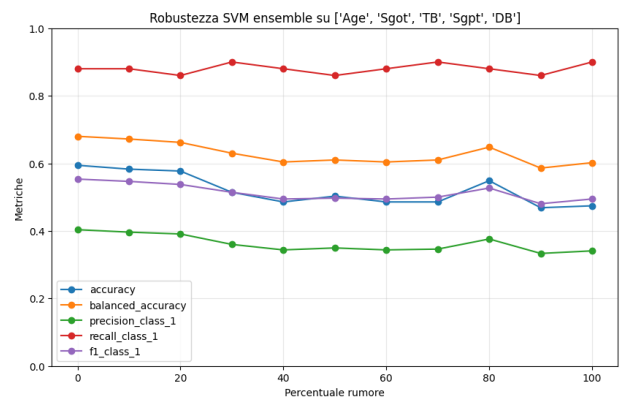


(b) Neural Network, feature migliori

Figura 3: Robustezza Neural Network – Liver.



(a) SVM, feature peggiori



(b) SVM, feature migliori

Figura 4: Robustezza SVM – Liver.

2 teenDataset.ipynb

2.1 Tabelle

Tabella 3: Mutual information raggruppata – Teen

Variabile	Importanza
anxiety_level	0.028224
daily_social_media_hours	0.026792
stress_level	0.020370
sleep_hours	0.016009
age	0.013230
gender	0.002635
social_interaction_level	0.000813
academic_performance	0.000720
platform_usage	0.000327
screen_time_before_sleep	0.000000
physical_activity	0.000000
addiction_level	0.000000

Tabella 4: Riepilogo Decision Tree – Teen

Gruppo	Score	Acc 0	Bal 0	Score 0	Acc 50	Bal 50	Score 50	Acc 100	Bal 100	Score 100
Peggiori	Macro-F1	0.9861	0.7222	0.9271	0.9861	0.7222	0.9271	0.9861	0.7222	0.9271
Migliori	Macro-F1	0.9861	0.7222	0.9271	0.9833	0.6667	0.8303	0.9778	0.5556	0.8720

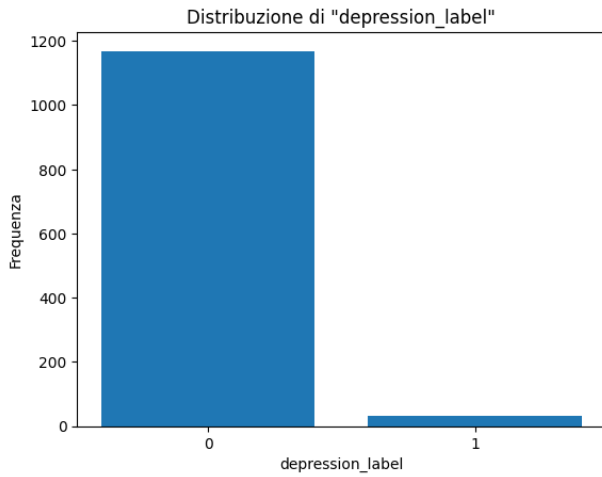
Tabella 5: Riepilogo Neural Network – Teen

Gruppo	Score	Acc 0	Bal 0	Score 0	Acc 50	Bal 50	Score 50	Acc 100	Bal 100	Score 100
Peggiori	F1 classe 1	0.9783	0.7832	0.5730	0.9733	0.7590	0.5033	0.9756	0.8251	0.5883
Migliori	F1 classe 1	0.9783	0.7832	0.5730	0.9772	0.7718	0.5484	0.9772	0.7826	0.5561

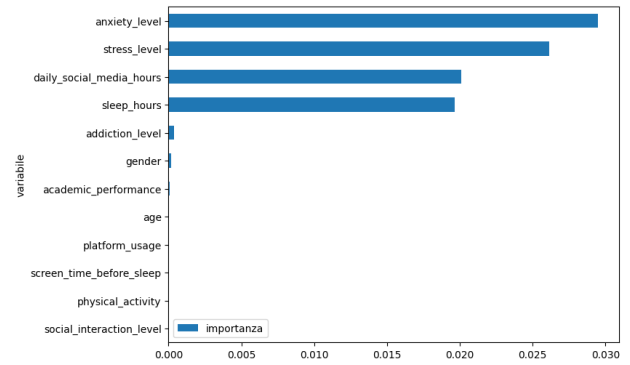
Tabella 6: Riepilogo SVM – Teen

Gruppo	Score	Acc 0	Bal 0	Score 0	Acc 50	Bal 50	Score 50	Acc 100	Bal 100	Score 100
Peggiori	F1 classe 1	0.9750	0.7165	0.4706	0.9806	0.7194	0.5333	0.9806	0.8818	0.6667
Migliori	F1 classe 1	0.9750	0.7165	0.4706	0.9806	0.7194	0.5333	0.9833	0.7749	0.6250

2.2 Grafici

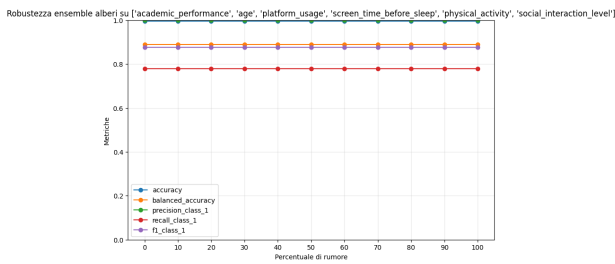


(a) Distribuzione del target

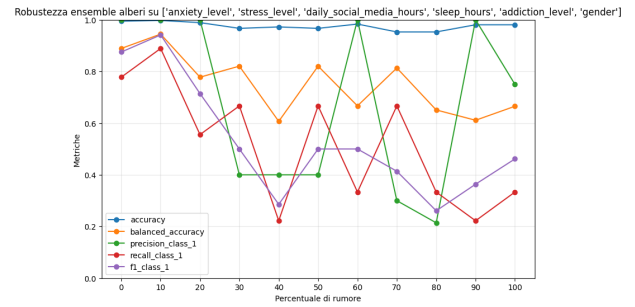


(b) Mutual information

Figura 5: Panoramica dataset Teen.

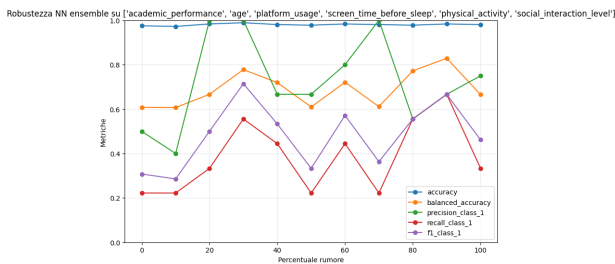


(a) Decision Tree, feature peggiori

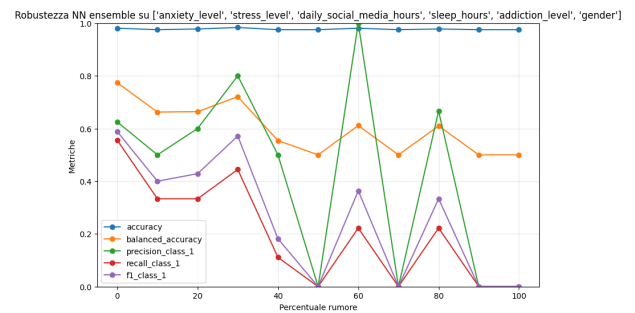


(b) Decision Tree, feature migliori

Figura 6: Robustezza Decision Tree – Teen.

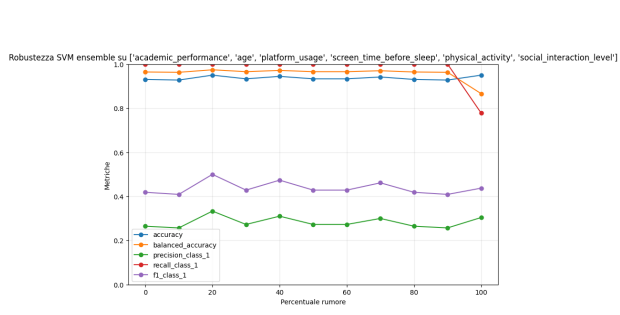


(a) Neural Network, feature peggiori

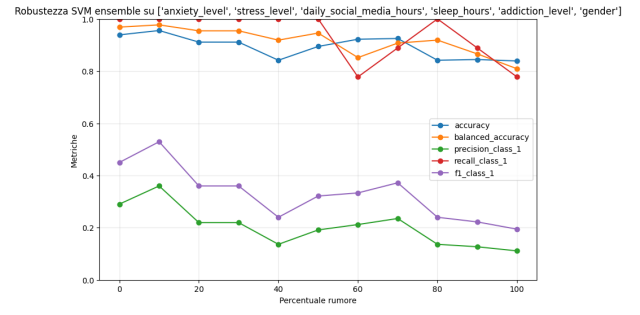


(b) Neural Network, feature migliori

Figura 7: Robustezza Neural Network – Teen.



(a) SVM, feature peggiori



(b) SVM, feature migliori

Figura 8: Robustezza SVM – Teen.

3 AdultIncomeDataset.ipynb

3.1 Tabelle

Tabella 7: Mutual information raggruppata – Adult

Variabile	Importanza
relationship	0.115938
marital-status	0.110479
capital-gain	0.080447
educational-num	0.066044
age	0.065317
occupation	0.063531
hours-per-week	0.042329
capital-loss	0.037595
fnlwgt	0.035118
gender	0.025859
workclass	0.015551
race	0.005985
native-country	0.005968

Tabella 8: Riepilogo SVM – Adult

Gruppo	Score	Acc 0	Bal 0	Score 0	Acc 50	Bal 50	Score 50	Acc 100	Bal 100	Score 100
Peggiori	F1 classe 1	0.7999	0.6282	0.4169	0.7989	0.5955	0.3282	0.7989	0.5955	0.3282
Migliori	F1 classe 1	0.7999	0.6282	0.4169	0.8082	0.6453	0.4536	0.7732	0.5883	0.3302

Nota: nel file `AdultIncomeDataset.ipynb` i grafici finali per Decision Tree e Neural Network sono presenti, ma nel notebook salvato non risultano serializzati gli output tabellari finali delle stesse prove. Ho quindi incluso tutti i grafici esportati e le tabelle numeriche disponibili (Mutual Information e SVM).

3.2 Grafici

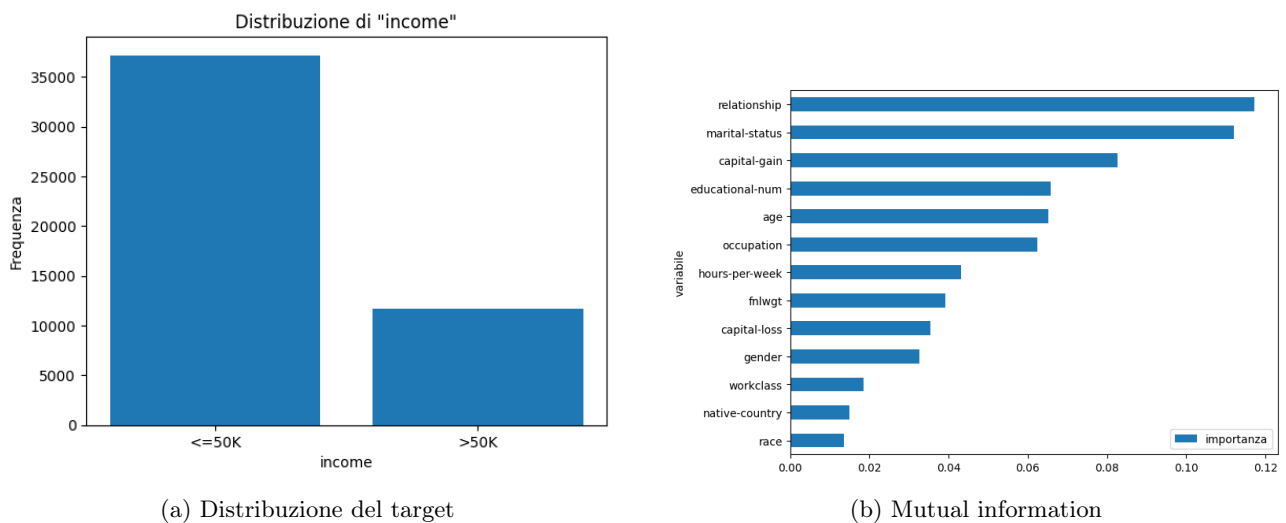
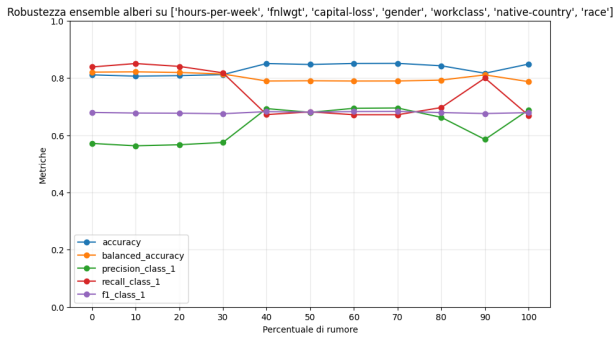
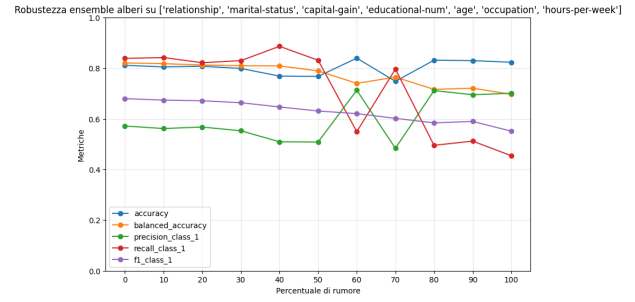


Figura 9: Panoramica dataset Adult Income.

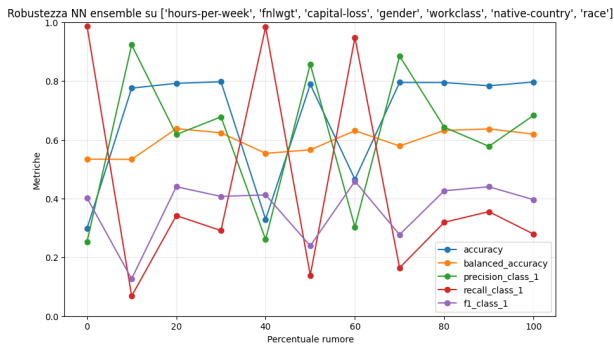


(a) Decision Tree, feature peggiori

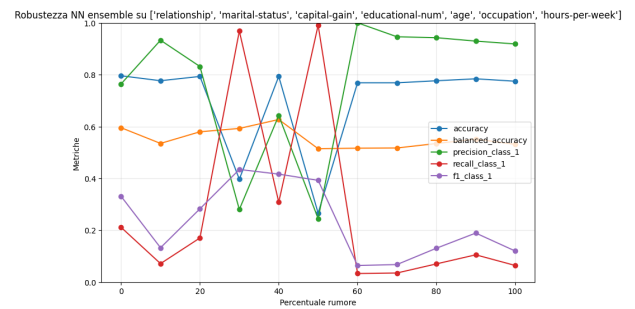


(b) Decision Tree, feature migliori

Figura 10: Robustezza Decision Tree – Adult Income.

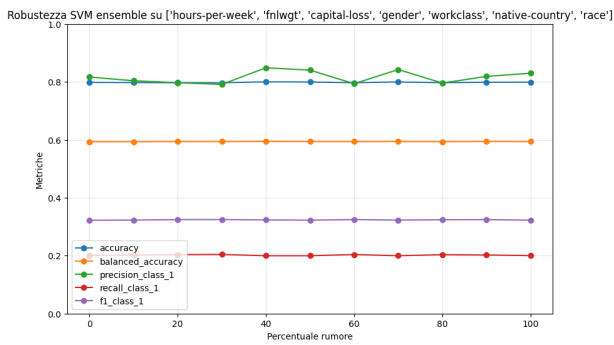


(a) Neural Network, feature peggiori

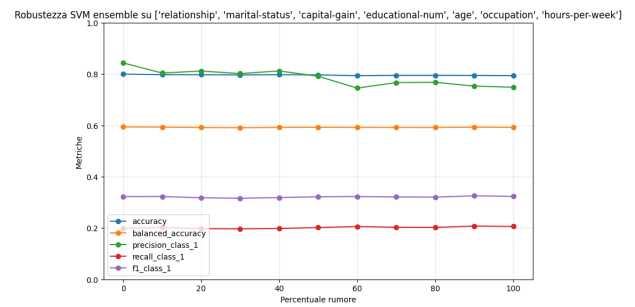


(b) Neural Network, feature migliori

Figura 11: Robustezza Neural Network – Adult Income.



(a) SVM, feature peggiori



(b) SVM, feature migliori

Figura 12: Robustezza SVM – Adult Income.